

Customer Retention Enhancement Through Predictive Analytics

1. Introduction

Customer churn is a major challenge for banks because losing customers can reduce revenue and market competitiveness. The objective of this project is to analyse customer data, identify factors influencing churn, and prepare a clean dataset for predictive modelling. This report presents the data gathering process, exploratory data analysis (EDA), and data preprocessing activities performed on the customer churn dataset.

2. Data Gathering

Dataset Selection

The customer churn dataset contains information related to customer demographics, spending behaviour, service usage, and churn status.

Features Used

- Customer ID
- Gender
- Income Level
- Amount Spent
- Login Frequency
- Service Usage
- Resolution Status
- Churn Status

Rationale for Selection

These variables were selected because they provide valuable information about customer behaviour and engagement. Spending patterns, login activity, service usage, and customer demographics are commonly associated with customer retention and churn behaviour.

3. Exploratory Data Analysis (EDA)

EDA was conducted to understand the structure of the dataset, identify patterns, and explore relationships between variables

The EDA process provided valuable insights into customer behaviour, engagement levels, and transaction patterns. These findings helped identify important features for machine learning model development and guided the data preprocessing steps required to prepare the dataset for predictive analytics.

The visualisations and statistical summaries generated during this stage are presented in the following sections of the report.

.

DATASET INFORMATION

```
print(df_merged.info())
```

```
.. <class 'pandas.core.frame.DataFrame'>
RangeIndex: 6812 entries, 0 to 6811
Data columns (total 14 columns):
 #   Column              Non-Null Count  Dtype  
---  -
 0   CustomerID          6812 non-null   int64  
 1   Age                 6812 non-null   int64  
 2   Gender              6812 non-null   object  
 3   MaritalStatus       6812 non-null   object  
 4   IncomeLevel          6812 non-null   object  
 5   ChurnStatus          6812 non-null   int64  
 6   LastLoginDate        6812 non-null   datetime64[ns]
 7   LoginFrequency       6812 non-null   float64  
 8   ServiceUsage         6812 non-null   int64  
 9   InteractionID         5204 non-null   float64  
10   InteractionDate       5204 non-null   datetime64[ns]
11   InteractionType       5204 non-null   object  
12   ResolutionStatus     6812 non-null   int64  
13   AmountSpent          6812 non-null   float64  
dtypes: datetime64[ns](2), float64(3), int64(5), object(4)
memory usage: 745.2+ KB
None
```

STATISTICAL SUMMARY

```
print(df_merged.describe())
```

```
..
```

	CustomerID	Age	ChurnStatus	LastLoginDate \
count	6812.000000	6812.000000	6812.000000	6812
mean	500.169260	43.274516	0.198473	2023-07-06 04:56:47.633587712
min	1.000000	18.000000	0.000000	2023-01-01 00:00:00
25%	247.000000	30.000000	0.000000	2023-04-12 00:00:00
50%	505.500000	44.000000	0.000000	2023-07-13 00:00:00
75%	750.000000	56.000000	0.000000	2023-10-01 00:00:00
max	1000.000000	69.000000	1.000000	2023-12-31 00:00:00
std	286.704642	15.286788	0.398880	NaN

	LoginFrequency	ServiceUsage	InteractionID \
count	6.812000e+03	6812.000000	5204.000000
mean	4.485223e-17	0.977540	5921.861261
min	-1.758361e+00	0.000000	2015.000000
25%	-8.338177e-01	0.000000	3873.250000
50%	1.960671e-02	1.000000	5903.000000
75%	8.730311e-01	2.000000	7851.000000
max	1.655337e+00	2.000000	9997.000000
std	1.000073e+00	0.802796	2332.331260

	InteractionDate	ResolutionStatus	AmountSpent
count	5204	6812.000000	6.812000e+03
mean	2022-07-01 16:31:10.561106944	0.840722	2.086150e-18
min	2022-01-01 00:00:00	0.000000	-1.724672e+00
25%	2022-04-07 00:00:00	0.000000	-8.714359e-01
50%	2022-07-02 00:00:00	1.000000	1.570936e-03
75%	2022-09-25 00:00:00	1.000000	8.654100e-01
max	2022-12-30 00:00:00	2.000000	1.737262e+00
std	NaN	0.778528	1.000073e+00

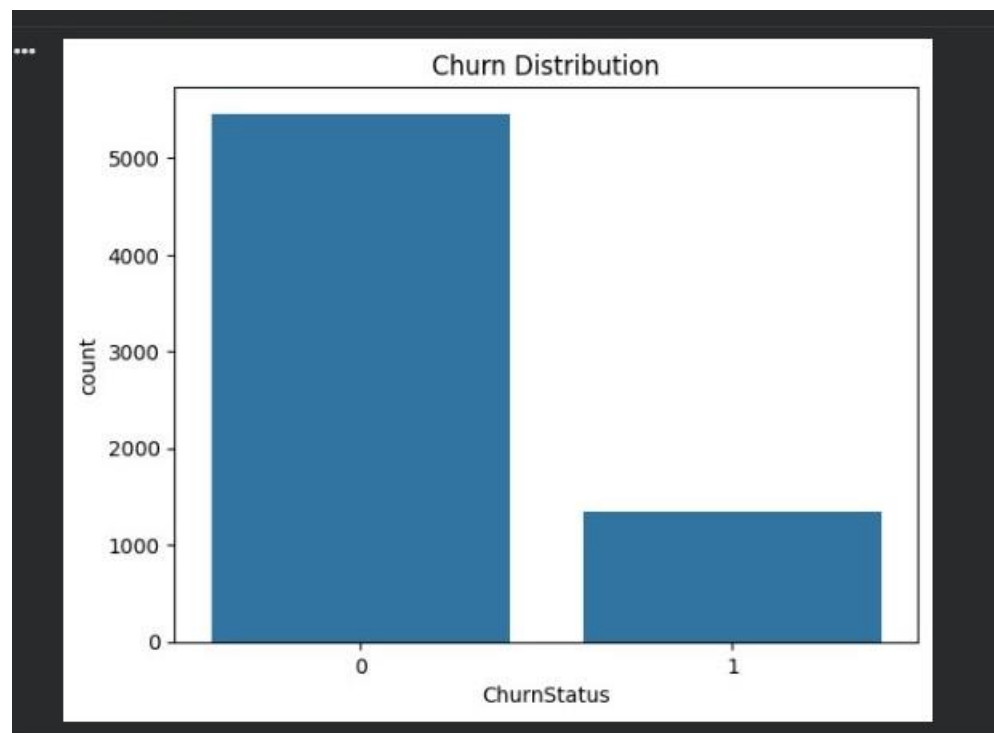
MISSING VALUES ANALYSIS(EDA)

```
print(df_merged.isnull().sum())
```

```
*** CustomerID      0
    Age            0
    Gender          0
    MaritalStatus   0
    IncomeLevel     0
    ChurnStatus     0
    LastLoginDate   0
    LoginFrequency  0
    ServiceUsage    0
    InteractionID    1608
    InteractionDate  1608
    InteractionType  1608
    ResolutionStatus 0
    AmountSpent     0
    dtype: int64
```

VISUALISATIONS

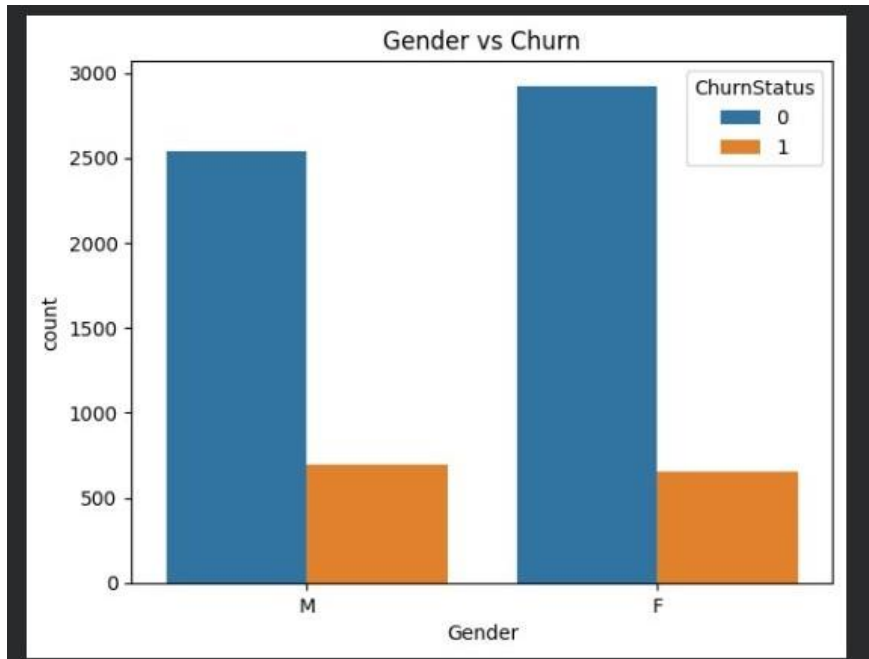
Churn Distribution



Observation:

The graph illustrates the distribution of customers who remained with the bank and those who churned.

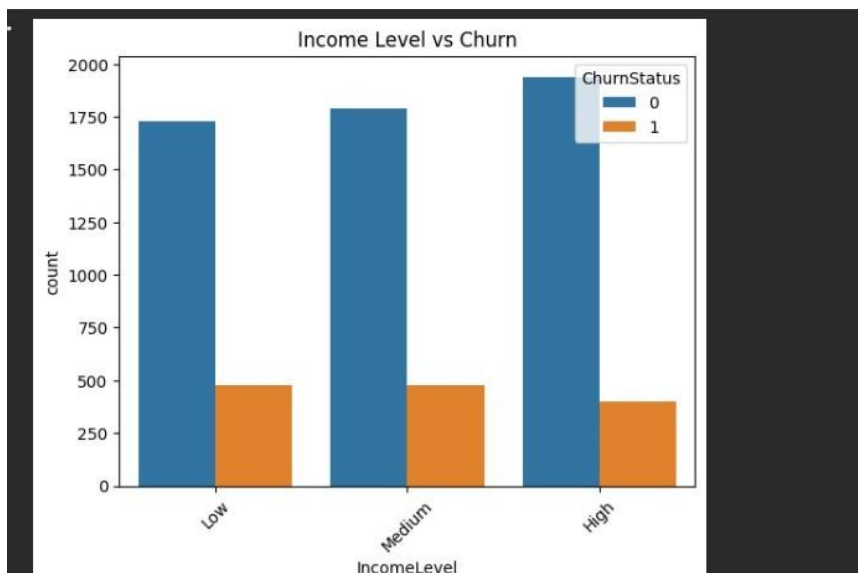
Gender vs Churn



Observation:

This analysis helps identify whether customer churn varies across gender groups.

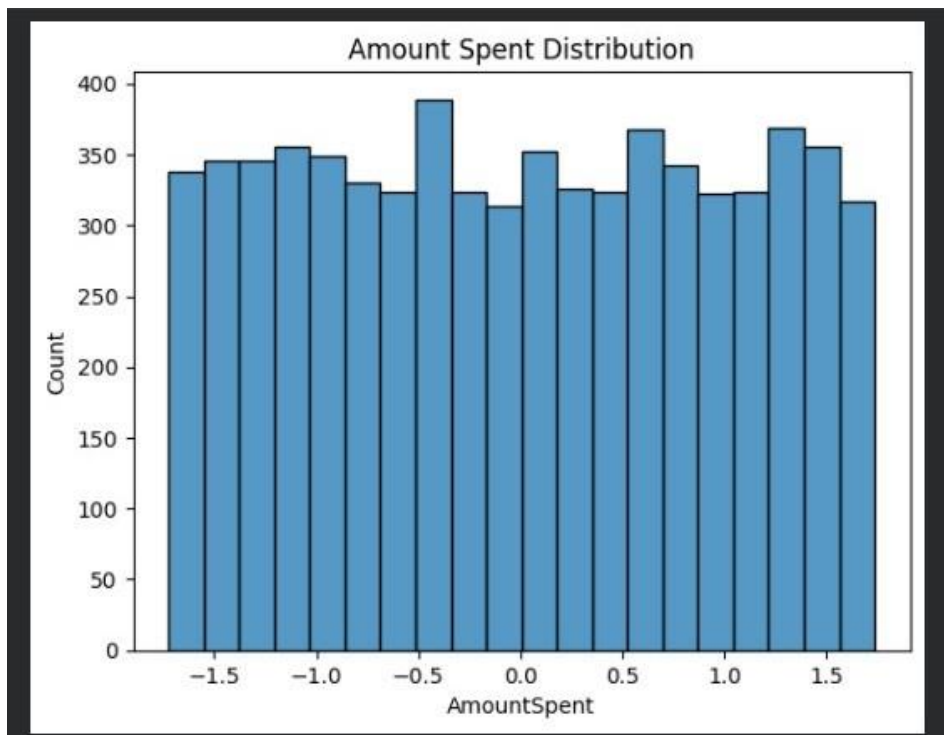
Income Level vs Churn



Observation:

Different income categories may exhibit different churn behaviours.

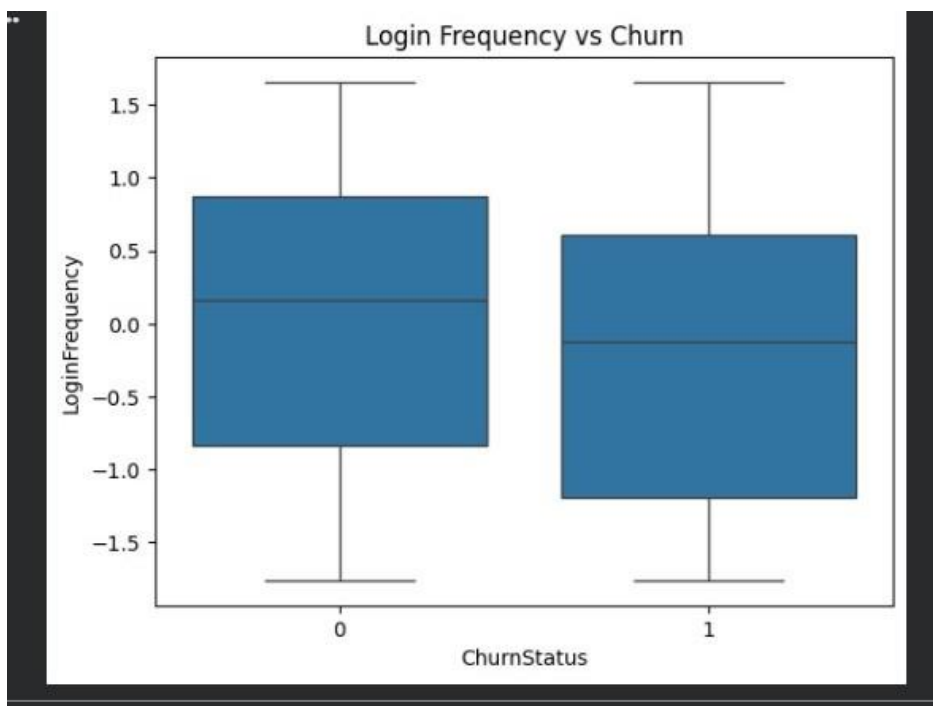
Amount Spent Distribution



Observation:

The spending behaviour of customers was analysed to understand transaction patterns.

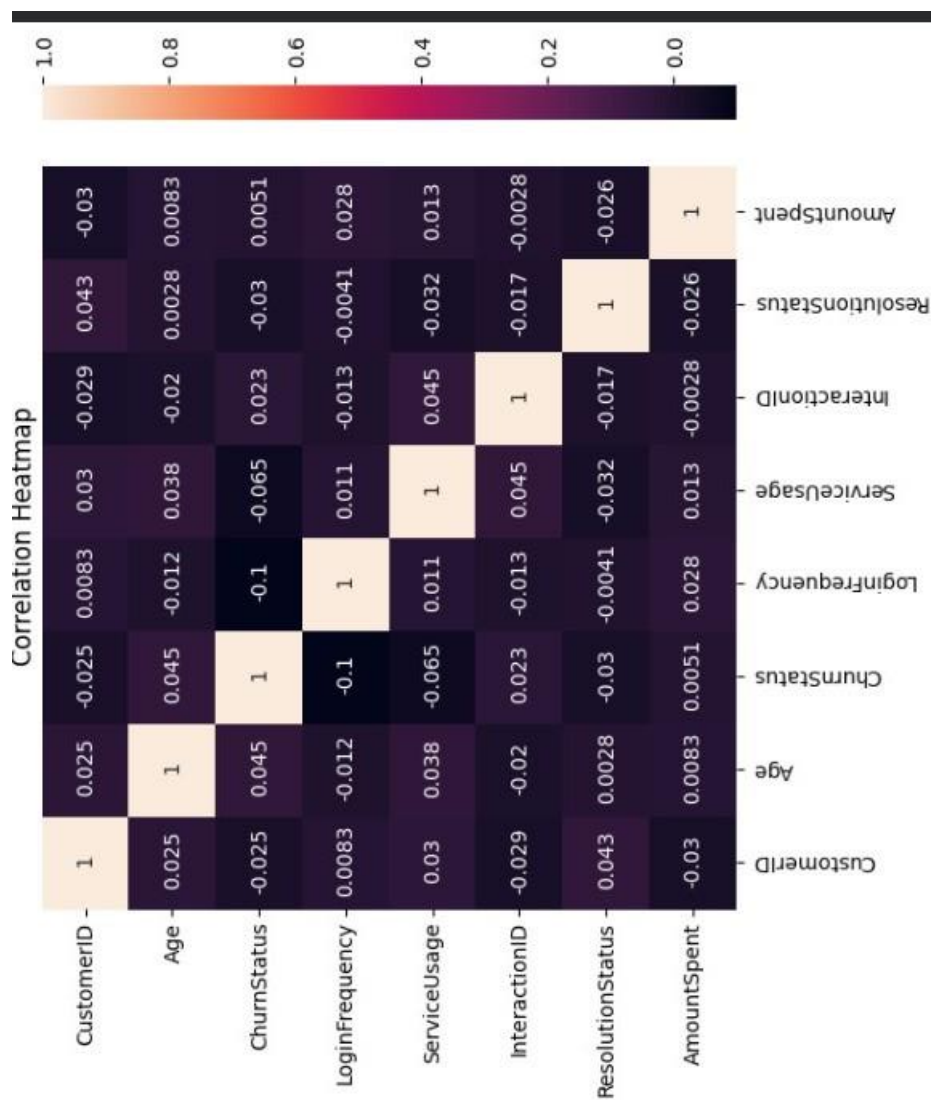
Login Frequency vs Churn



Observation:

Customer login activity was examined to determine its relationship with churn.

Correlation Heatmap



Observation:

The heatmap highlights relationships between numerical variables and their association with churn.

4. Data Cleaning and Preprocessing

The following preprocessing steps were performed:

Missing Value Handling

The dataset was checked for missing values and appropriate actions were taken where necessary.

Duplicate Record Handling

Duplicate records were identified and removed to improve data quality.

Encoding Categorical Variables

Categorical variables such as Gender, Income Level, Service Usage, and Resolution Status were converted into numerical representations using encoding techniques.

Feature Scaling

Numerical variables such as Amount Spent and Login Frequency were standardised to ensure consistency across features.

5. Cleaned Dataset

After preprocessing, the dataset was transformed into a machine-learning-ready format. The cleaned dataset was saved as:

Cleaned_Customer_Churn_Data.xlsx

This dataset will be used in the next phase for predictive model development.

6. Conclusion

The dataset was successfully collected, analysed, cleaned, and prepared for machine learning. Exploratory analysis provided insights into customer behaviour and potential churn indicators. The final preprocessed dataset is suitable for building predictive models aimed at identifying customers at risk of churn.